

Day 5 — Adraka (Ginger)

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can identify *adraka* (*Zingiber officinale*) as a rhizome, distinguish *adraka* (fresh) from *sūṇṭhī* (dried) and explain that the active compound shifts from **gingerol** to **shogaol** on drying, and describe ginger’s traditional classification as *uṣṇa* (heating) with one well-supported modern use.

Materials

- Fresh ginger rhizome
- Small jar of dried ground ginger powder (*sūṇṭhī*)
- Optional: a thumb-size lump of jaggery for the heating-demo
- Optional, for demo: a small kettle, ginger slices, water, lemon, jaggery — ginger tea
- Whiteboard

Vocabulary introduced today

- *adraka* (IAST: *adraka*; lit. “ginger”) — fresh ginger rhizome.
- *sūṇṭhī* (IAST: *sūṇṭhī*; lit. “dried ginger”) — same plant, dried rhizome.
- **Gingerol** — the principal pungent compound in fresh ginger ([6]-gingerol).
- **Shogaol** — formed when gingerol loses a water molecule during drying or heating. More pungent. The dominant pungent compound in dried ginger.

Lesson flow

Warm-up (5 min)

Hold up the fresh ginger and the dried ginger powder.

“*Same plant. Same rhizome. One fresh, one dried. Which one will be more pungent?*” Take a vote.

Pass tiny crumb of dried ginger and a tiny shaving of fresh ginger to each student (separate plates). Smell only. Discuss.

Core (20 min)

1. **Identify it.** Like turmeric, ginger is a **rhizome** — underground stem. The two plants are related (same family: Zingiberaceae). Differences:
 - **Ginger:** pale tan skin; pale yellow-white interior; spicy-pungent smell.
 - **Turmeric:** pale brown skin; bright orange interior; earthy-pepper smell.
 - **Leaves above ground:** very similar — long, blade-like, alternating up a stem.
2. **The fresh-vs-dried switch.** This is the chemistry of the day.

- **Fresh ginger** is dominated by **[6]-gingerol** — a phenolic compound with a long carbon side-chain ending in a hydroxyl (–OH).
- When ginger is **dried** (loses water content) or heated, gingerol undergoes a dehydration reaction. The hydroxyl group is lost as water, leaving **[6]-shogaol**.
- Shogaol is more pungent than gingerol — it's why dried ginger powder is sharper on the tongue than fresh ginger.
- In Ayurveda, this same transformation is noticed: *adraka* (fresh) is described as somewhat less heating than *sūnṭhī* (dried). The classical text and the modern molecule agree.

Sketch on board:

gingerol → (drying / heating) → shogaol + H₂O
(fresh) (dried, more pungent)

3. What ginger does — well-supported uses.

- **Anti-nausea** — one of the few traditional herbs with **multiple positive clinical trials**, especially for:
 - Pregnancy-related nausea (under doctor supervision — pregnancy dosing is not a school question).
 - Motion sickness — small ginger doses (1–2 g, e.g. a chewed slice) have outperformed placebo in some trials.
 - Post-surgical nausea — used in some hospital protocols.
- **Warming feeling** — ginger does cause a measurable rise in skin and body temperature (“thermogenic” effect). Activates TRPV1 receptors — the same receptor activated by capsaicin in chillies.
- **Digestive support** — culinary use after meals (a slice in tea, or ginger pickle) has a long tradition; mechanism includes stimulation of saliva and gastric secretions.

4. Traditional Ayurvedic frame. *Adraka* is *kaṭu* (pungent) in taste, *uṣṇa* (heating) in *vīrya*. Used in:

- Tea / decoction for cold, congestion, light fever (small culinary amounts).
- Combined with jaggery for a sore throat (folk remedy; some mechanistic justification through the soothing/warming pairing).

5. Cautions:

- Large doses can cause heartburn or stomach upset.
- Pregnancy — doses higher than culinary need supervision.
- Some blood-thinning medications (warfarin) interact with high ginger intake. Not a school-dose concern but worth knowing.

Activity (15 min) — Ginger tea demo + Formative Quiz Part B

First 7 min — Tea demo: - Teacher slices a 1 cm piece of fresh ginger, drops into 200 ml hot water, lets steep 3–4 minutes. - Optional: add a tiny pinch of jaggery and a drop of lemon. - Pour into a clear cup. Pass around so students can see the colour (very pale yellow) and smell the

aroma (sharp, pungent, citrus-warm). - Discussion: which compound dominates here — gingerol or shogaol? (Since the ginger is fresh and only briefly heated: mostly gingerol.) If the ginger had been dried first and then steeped: more shogaol.

Next 8 min — Formative Quiz Part B (from quizzes/formative.md). Hand out the 10-question slip. Pencils only. Collect at end of class.

Wrap-up (5 min)

- Restate safety rule + the anti-nausea note (“ginger has more clinical evidence than most herbs, but for *specific* uses”).
- Preview Day 6: “*Tomorrow we meet a seed that takes the digestive role even further — ajwain.*”

Student-facing content

Adraka (*Zingiber officinale*) — known in English as ginger.

What it is: A **rhizome** (underground stem), related to turmeric. Pale tan outside, pale yellow inside.

Fresh vs dried — same plant, different chemistry:

Form	Sanskrit name	Dominant compound
Fresh rhizome	<i>adraka</i>	gingerol
Dried rhizome	<i>sūṇṭhī</i>	shogaol (more pungent)

When ginger dries or heats, gingerol loses a water molecule and becomes shogaol.

Well-supported modern uses: - Anti-nausea, especially motion sickness and pregnancy-related (under medical supervision). - Mild warming effect (activates TRPV1 receptors — same as chilli’s capsaicin). - Digestive support in culinary amounts.

Ayurvedic classification: *kaṭu* (pungent) in taste, *uṣṇa* (heating) in *vīrya*.

Cautions: Large doses → heartburn or stomach upset; pregnancy doses need a doctor; interactions with blood-thinning medications.

Homework

- Watch what ginger is used in your home over the next two days. List 3 dishes / drinks that contain it. For each, guess: fresh or dried? Heating or just flavour?

Differentiation

- **One tier up:** Look up the chemical structure of gingerol and shogaol. Identify the difference (one OH group + one double bond gain). Sketch both in your notebook.
- **One tier down:** Focus on identifying ginger by sight and smell, and remembering “fresh = gingerol, dried = shogaol.”

Teacher notes

- The chemistry sketch (gingerol → shogaol) is the headline. If a student remembers nothing else from today, they should know that drying ginger changes its chemistry, and the change is well-defined.
- The “ginger has real clinical evidence for nausea” point is important — students often don’t realise that for *some specific herbal uses*, the clinical evidence is genuinely strong. It builds the right kind of nuance: not blanket scepticism, not blanket endorsement, but herb-by-herb, use-by-use evaluation.
- If a student in your class is pregnant (older Middle band, occasionally) or has a pregnant family member raising questions, redirect gently: clinical dosing is for clinicians.

Citation(s) used in this lesson

- Gingerol → shogaol dehydration on drying: standard pharmacognosy textbook content.
- Ginger as *adraka* (fresh) and *sūnṭhī* (dried) — distinguished in *Caraka Saṁhitā* and standard Ayurvedic nighantus (lexicons). Chapter-level reference only.
- Anti-nausea clinical evidence: multiple systematic reviews exist; teachers can search “ginger nausea systematic review” in PubMed.